E-commerce analysis: where the profit comes from

Georgian online store, Jan 2024 - Sep 2026. Synthetic dataset. Delivered orders only; profit is after product cost and refunds.

1,810,324 GEL	495,633 GEL	8,485	4,943
Net revenue	Profit after returns	Delivered orders	Buying customers

What to do next

1. Fashion: return rate 13.0% (next highest: 7.6%), margin after returns 19.1%. All 160 'Wrong size' returns are in Fashion. Add size guides and fit notes before adding more stock.

2. Acquisition: customers from Instagram/Facebook Ads bring 296 GEL on average vs 380 GEL from Referral. Shift part of the ads budget toward referral and email.

3. Discounts: BF25 and SUMMER15 earn 8.5% and 16.1% margin after returns, against 27.5% for orders without a code. Cap the depth of these campaigns.

4. Retention: 56% of buyers purchase only once, and repeat orders come about 52 days apart. A reminder email around day 45 after each order is the first test to run.

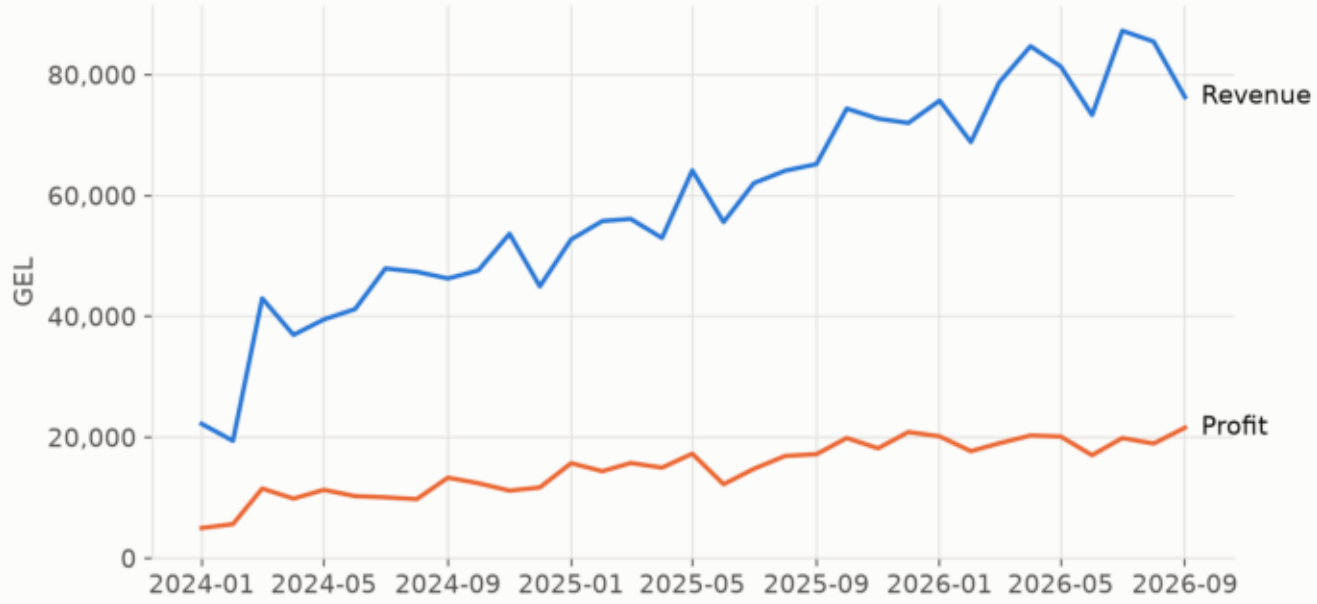
5. Assortment: Zentra Backpack Lite (P0081) lost money after returns (-580 GEL). Review its price, cost and returns.

Tools: SQL (SQLite), Python (pandas, matplotlib). Full code in the project repository.

Growth and profitability

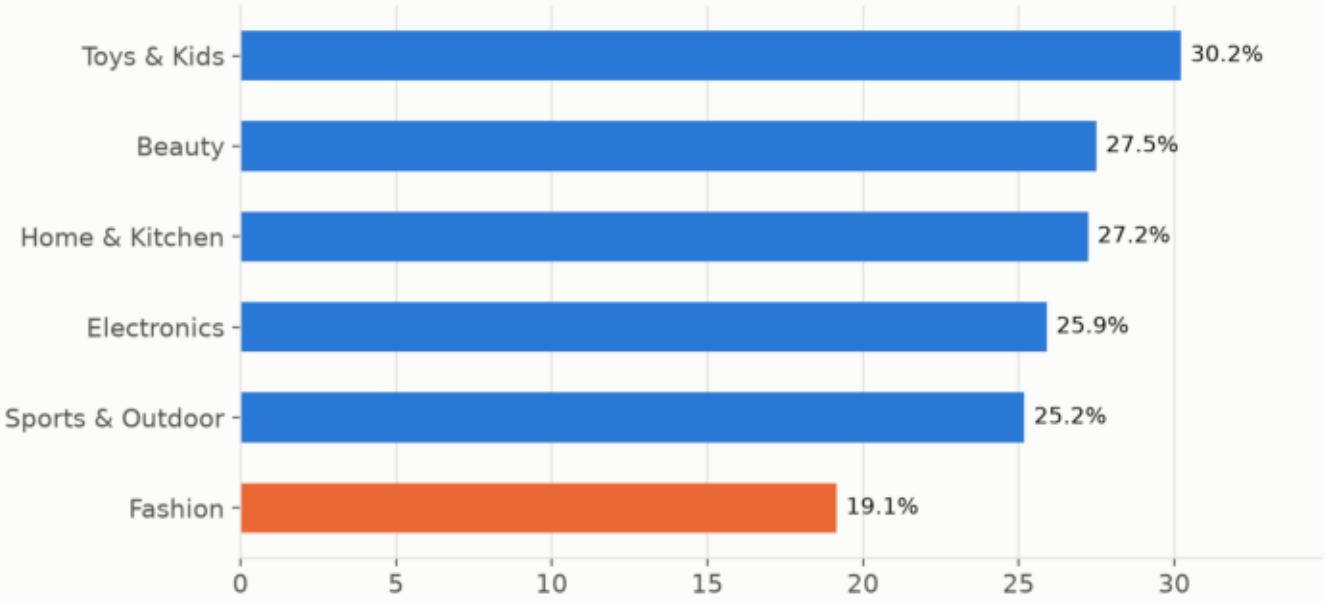
Revenue and profit have both nearly doubled

Last 12 months vs first 12 months: revenue +90%, profit +91%. Delivered orders, profit after returns.



Fashion keeps the least profit per lari of revenue

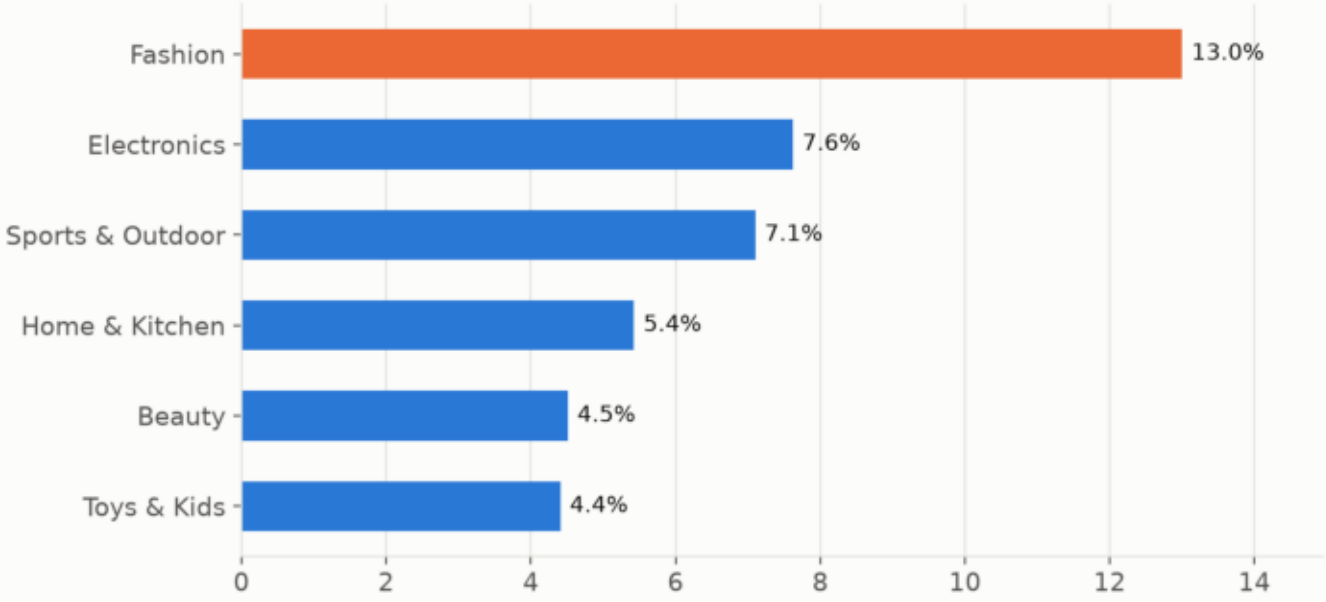
Profit margin after returns, by category



Returns and customers

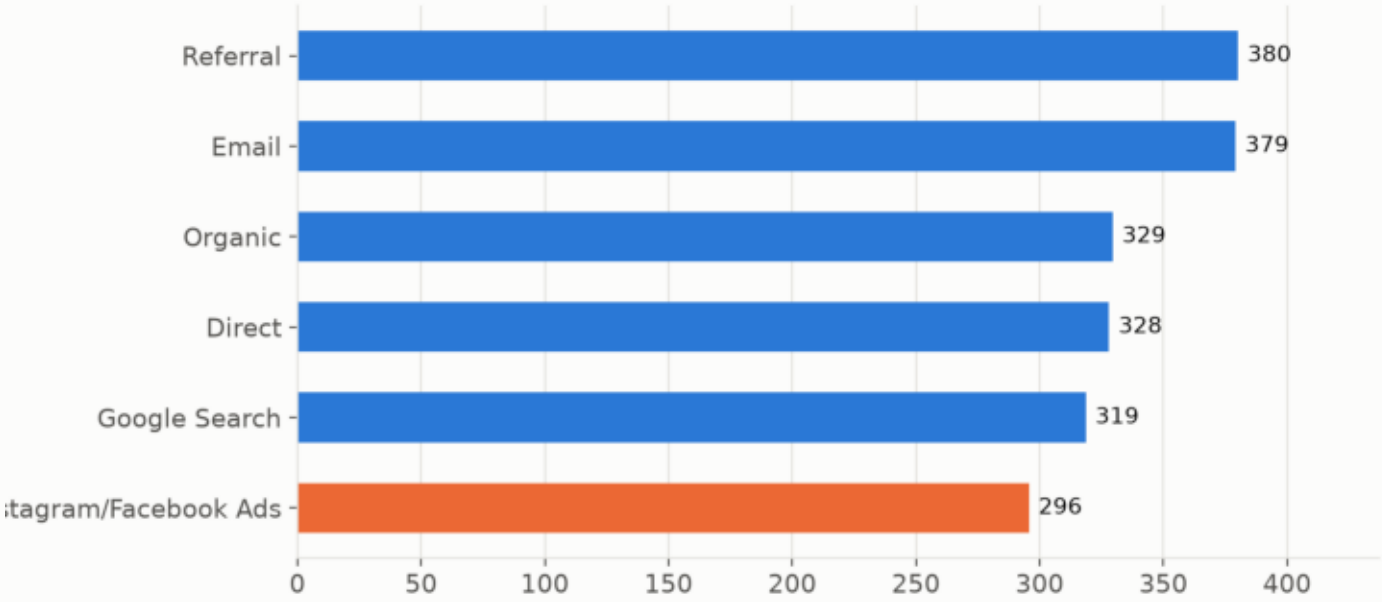
Fashion returns are 1.7x the next-highest category

Share of items sold that were returned



Customers from Instagram/Facebook Ads spend the least

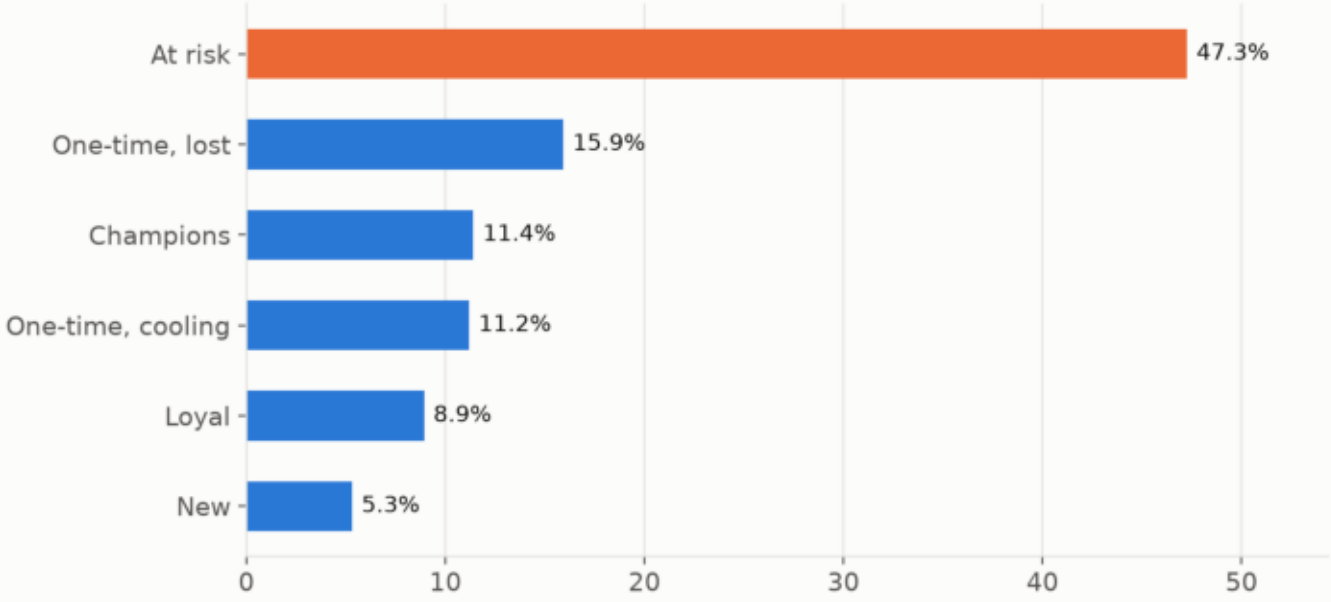
Average revenue per acquired customer, GEL (delivered orders)



Loyalty

'At risk' customers hold 47% of revenue

Share of net revenue by customer segment (2+ orders, last purchase over 180 days ago = At risk)



Fewer than 1 in 5 new customers buys again the next month

Monthly cohort retention. Average month-1 retention is 18% and fades steadily after that.

